

EXPERIMENT: 26

TO IMPLEMENT DATE AND TIME DISPLAY FROM CLIENT TO SERVER USING TCP SOCKETS IN JAVA/C

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <time.h>
#include <sys/wait.h>

int main()
{
    int server, client, sock;
    struct sockaddr_in addr;
    char buffer[100];
    time_t t;
    pid_t pid;
    pid = fork();
    if (pid == 0)
    {
        server = socket(AF_INET, SOCK_STREAM, 0);
        addr.sin_family = AF_INET;
        addr.sin_port = htons(8080);
        addr.sin_addr.s_addr = INADDR_ANY;
        bind(server, (struct sockaddr *)&addr, sizeof(addr));

        listen(server, 1);

        printf("Server waiting...\n");

        client = accept(server, NULL, NULL);
```

```

t = time(NULL);
strcpy(buffer, ctime(&t));
send(client, buffer, strlen(buffer), 0);
recv(client, buffer, sizeof(buffer), 0);
buffer[sizeof(buffer) - 1] = '\0';
printf("Client IP Address: %s\n", buffer);
printf("Date and time sent successfully.\n");
close(client);
close(server);
}
else
{
sleep(1);

sock = socket(AF_INET, SOCK_STREAM, 0);

addr.sin_family = AF_INET;
addr.sin_port = htons(8080);
addr.sin_addr.s_addr = inet_addr("127.0.0.1");

connect(sock, (struct sockaddr *)&addr, sizeof(addr));

strcpy(buffer, "127.0.0.1");

send(sock, buffer, strlen(buffer), 0);

recv(sock, buffer, sizeof(buffer), 0);
buffer[sizeof(buffer) - 1] = '\0';

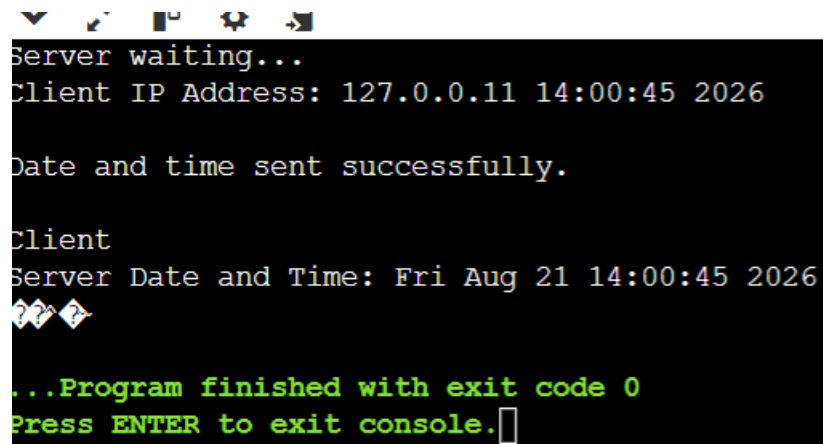
printf("\nClient\n");
printf("Server Date and Time: %s", buffer);

```

```
        close(sock);

        wait(NULL);
    }

    return 0;
}
```



```
Server waiting...
Client IP Address: 127.0.0.11 14:00:45 2026

Date and time sent successfully.

Client
Server Date and Time: Fri Aug 21 14:00:45 2026
...Program finished with exit code 0
Press ENTER to exit console.
```